

Name: _____

Student ID: _____

Question:	1	2	3	4	5	6	7	Total
Points:	7	8	2	4	7	5	3	36
Score:								

Grade: _____

$$\text{Grade} = \frac{\# \text{total points}}{4} + 1$$

Read the instructions carefully!

- The use of a calculator, book or notes is **not** allowed.
 - Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page, but make sure the final answer starts on the front.
 - You can ask for extra paper to do your calculations, but you shall not submit it, the graded answers will be the ones on these papers, so make sure the final answers are properly explained here, starting in the blank space provided and finishing on the back if necessary.
 - Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for full score.
 - If you feel like you will be short in time, prioritize questions you know or the ones with higher score count.
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1. Let $\mathbf{x} = (x_1, x_2, x_3, x_4)$ and $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation given by

$$T(\mathbf{x}) = \begin{bmatrix} x_1 + 4x_2 + 5x_3 + 2x_4 \\ 2x_1 + x_2 + 3x_3 \\ -x_1 + 3x_2 + 2x_3 + 2x_4 \end{bmatrix}.$$

(a) **(1 pts.)** Is the map linear? Why?

(b) **(3 pts.)** Find $\ker T$.

Hint: Provide a basis for $\ker T = \{\mathbf{x} \in \mathbb{R}^4 : T(\mathbf{x}) = 0\}$.

(c) **(3 pts.)** Find the range of T .

Hint: Provide a basis for $\text{range}(T) = \{T(\mathbf{x}) \in \mathbb{R}^3 : \mathbf{x} \in \mathbb{R}^4\}$.

2. Let $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$, where $a, b, c, d \in \mathbb{R}$.

- (a) **(4 pts.)** Under certain conditions for a, b, c, d it is possible to find an LU-factorization of A without doing any row interchange (swapping two rows). Find such conditions and determine the LU-factorization of A .

- (b) **(1 pts.)** Under such conditions, is A invertible? Why?

- (c) **(3 pts.)** For $a = 1$, $b = 2$, $c = 3$ and $d = 4$ compute the inverse of A .

3. (2 pts.) Let

$$A_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2 & \cdots & 2 \\ 1 & 2 & 3 & \cdots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \cdots & n \end{bmatrix}$$

with $n > 1$ be a generalization of the matrix A from the previous exercise 2. Calculate $\det A_n$ and explain the computation process and the results used.

Note: Do not use induction on n , but rather calculate the determinant of A_n .

4. Let $H = \left\{ \begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{bmatrix} : x_1, x_3 \in \mathbb{R} \right\} \subset \mathbb{R}^4$.

(a) **(2 pts.)** Show that H is a subspace of $\mathbb{R}^4$.

(b) **(1 pts.)** Find a basis for H .

(c) **(1 pts.)** Let $L = \left\{ \begin{bmatrix} x_1 \\ 1 \\ x_3 \\ 1 \end{bmatrix} : x_1, x_3 \in \mathbb{R} \right\} \subset \mathbb{R}^4$. Is L a subspace of $\mathbb{R}^4$? Why?

5. Let

$$A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & 4 & 9 \\ 0 & 0 & 3 \end{bmatrix}.$$

(a) **(2 pts.)** Find all the eigenvalues of A . Can we know from this if A is diagonalizable?

(b) **(3 pts.)** Give a basis for every eigenspace of A .

(c) **(2 pts.)** Determine whether A is diagonalizable. If so, give the diagonalization, i.e., an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. If not, explain why A is not diagonalizable.

6. Let $W = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} \right\}.$

(a) **(4 pts.)** Find an orthonormal basis for W .

(b) **(1 pts.)** Calculate the distance between $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$ and W .

7. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

(a) **(1 pts.)** If T is a map between vector spaces and $T(0) = 0$, then T is linear.

(b) **(1 pts.)** If A and B are square matrices of the same size, then $\det(A + B) = \det A + \det B$.

(c) **(1 pts.)** Let A be a 4×7 matrix. It is possible for A to have rank 5 and nullity 2.

